

This paper develops a macro-finance asset pricing model for Treasury bonds and stock prices, based on Campbell-Cochrane (2005) habits and a number of other special features. Previous research (e.g., Piazzesi and Schneider, 2006, Rudebusch and Swanson, 2012, Campbell, Pflueger, and Viceira, 2020) has argued that when supply/inflation shocks dominate, the nominal bond term premium is positive, while when demand shocks dominate, the nominal bond term premium is negative. The present paper refines that argument by 1) focusing on the correlation between excess bond returns and excess stock returns, rather than the level of the term premium, and 2) pointing out that the monetary policy rule itself is important for this correlation, not just the relative importance of supply vs. demand shocks. The paper argues that, even in the presence of 1970s-80s-style supply shocks, the bond-stock correlation today remains mildly negative because the monetary policy rule is less reactive to inflation today than it was back in the 1980s.

Overall, my primary concern with the paper is that its main point and model are too specialized to be a good fit for a top general interest journal such as the AER. I also have some concerns about the details of the model, the fit of the model to the data, and the more recent very high correlation between bond and stock returns.

Main concerns

1. The macroeconomic model in the paper is fairly complicated and has a number of features that are likely to leave general audience readers baffled. Let us assume that the standard New Keynesian model and Campbell-Cochrane habits are relatively standard, although combining the two is not and is already starting to push a general audience. The model in the present paper then includes: 1) an output gap that is defined in terms of a trailing moving average of consumption (p. 7), 2) a bond preference shock ξ_t that affects bond interest rates but not expected stock returns (p. 8), 3) productivity that depends on recent economic activity x_t , as in Lucas' endogenous growth model (p. 10), and 4) partially adaptive subjective inflation expectations (p. 11), which affect households in the model but not the people who buy or sell bonds or stocks. This is really a quite complicated model with a lot of moving parts and a lot of assumptions that go into those moving parts. Even as someone is more specialized in this area, I am left with a number of questions and concerns about the model, which I describe further below. But this model is likely to be unintelligible to a general audience like that in the AER.

2. As mentioned above, some feature of the model puzzle and concern me. For example, the bond preference shock ξ_t affects only bond prices but not stock prices (p. 8). But that means that there should be ***two* Euler equations in the model and not just one (equation 17)**: an Euler equation for investing in stocks as well as the Euler equation (17) for investing in bonds. Importantly, the Euler equation for investing in stocks does not have a ξ_t shock in it because ξ_t only affects bond pricing and not stock pricing. This implies that shocks to ξ_t affect bond yields but do not affect consumption or output, because the stock-return Euler equation is unaffected. I am thus puzzled by the discussion at the end of Section 2.3 that says that shocks to ξ_t affect consumption and output. Thus, even someone who is relatively more specialized in the area is having problems figuring

out what is going on in this model.

3. Equation (5) defines the output gap x_t to be the deviation of consumption c_t from a long trailing moving average of past consumption. But then the sentence after equation (5) defines the output gap to be the difference between output and flex-price flex-wage output. I don't see how these two definitions agree—they seem to contradict each other. So here is another case in which the model is confusing even to someone more specialized in the field.

4. The fit of the model in Figure 2 and middle panel of Table 3 is not very good. This is problematic because these are the main empirical features of the data that the model is purporting to explain.

5. Finally, the paper makes a big deal that nominal bond-stock return betas have stayed negative (Figure 1 and pp. 1-2, p. 33). But almost immediately after the end of the sample period in the paper, the correlation between nominal bond returns and stock returns has skyrocketed—i.e., long-term bond yields and stock prices are moving almost exactly opposite to each other every day. I think the author could potentially increase the appeal of the paper by using the model to also explain this recent striking behavior, but this recent episode is currently omitted and seems at odds with the way the paper is being pitched currently.

Minor Comments

I also noticed a few more minor comments and suggestions as I read through the paper:

1. The assumption that households have partially adaptive subjective inflation expectations but the agents who trade stocks and bonds do not is somewhat problematic, because the latter asset trading agents are never defined in the model. I think for clarity these agents need to be explicitly defined.

2. I think of supply shocks as a 1970s phenomenon, not a 1980s phenomenon. Thus, a lot of the discussion in the paper about the nature of shocks in the 1980s seems a little misplaced.

3. After equation (2), risk aversion increases when consumption falls, not decreases.

4. Before equation (11), I wouldn't call the one-period bond rate the "risk-free rate", because the bond demand shock ξ_t makes the one-period bond rate different from the shadow risk-free rate.

5. At the end of Section 4.1, the idea that it is the *interaction* between monetary policy and supply shocks that is crucial for generating stagflation is essentially the same point made by Primiceri in his 2006 QJE paper.